

205 - Submitting Your First Job



Create a simple Slurm script, submit it with sbatch, monitor it with squeue, and read the resulting output file.

Directory Setup

1

Create a folder for your first job
Type "**mkdir -p ~/first-job-demo** Return"

```
pranavangadi — pangadi7@login-ice-gnr-2:~ — ssh pangadi7@login-ice....  
This computer system is the property of Georgia Institute of  
Technology. Any user of this system must comply with all Institute  
and Board of Regents policies, including the Acceptable Use  
Policy (AUP), Data Privacy Policy (DPP) and Cyber Security  
Policy (CSP), see http://b.gatech.edu/it-policies. Users should  
have no expectation of privacy, as any and all files on this system  
may be intercepted, monitored, copied, inspected, and/or disclosed to  
authorized personnel in order to meet institute obligations.  
  
By using this system, I acknowledge and consent to these terms.  
  
Software or services appearing on the State of Georgia's Prohibited  
Software & Services list may not be used on this computer system.  
This includes DeepSeek and Qwen (an Alibaba product).  
The full list may be found at  
https://gta-psg.georgia.gov/psg/prohibited-software-services-ss-22-002.  
  
#####  
# Welcome to GT Instructional Cluster #  
#####  
  
If you require assistance with this system, please contact your  
course instructor or teaching assistant (TA).  
[[pangadi7@login-ice-gnr-2 ~]$ mkdir -p ~/first-job-demo
```

2

Open the first-job folder
Type "**cd ~/first-job-demo** Return"

```
pranavangadi — pangadi7@login-ice-gnr-2:~ — ssh pangadi7@login-ice....  
Technology. Any user of this system must comply with all Institute  
and Board of Regents policies, including the Acceptable Use  
Policy (AUP), Data Privacy Policy (DPP) and Cyber Security  
Policy (CSP), see http://b.gatech.edu/it-policies. Users should  
have no expectation of privacy, as any and all files on this system  
may be intercepted, monitored, copied, inspected, and/or disclosed to  
authorized personnel in order to meet institute obligations.  
  
By using this system, I acknowledge and consent to these terms.  
  
Software or services appearing on the State of Georgia's Prohibited  
Software & Services list may not be used on this computer system.  
This includes DeepSeek and Qwen (an Alibaba product).  
The full list may be found at  
https://gta-psg.georgia.gov/psg/prohibited-software-services-ss-22-002.  
  
#####  
# Welcome to GT Instructional Cluster #  
#####  
  
If you require assistance with this system, please contact your  
course instructor or teaching assistant (TA).  
[[pangadi7@login-ice-gnr-2 ~]$ mkdir -p ~/first-job-demo  
[[pangadi7@login-ice-gnr-2 ~]$ cd ~/first-job-demo
```

3

Confirm your working directory
Type "**pwd** Return"

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...

Policy (CSP), see http://b.gatech.edu/it-policies. Users should
have no expectation of privacy, as any and all files on this system
may be intercepted, monitored, copied, inspected, and/or disclosed to
authorized personnel in order to meet institute obligations.

By using this system, I acknowledge and consent to these terms.

Software or services appearing on the State of Georgia's Prohibited
Software & Services list may not be used on this computer system.
This includes DeepSeek and Qwen (an Alibaba product).
The full list may be found at
https://gta-psg.georgia.gov/psg/prohibited-software-services-ss-22-002.

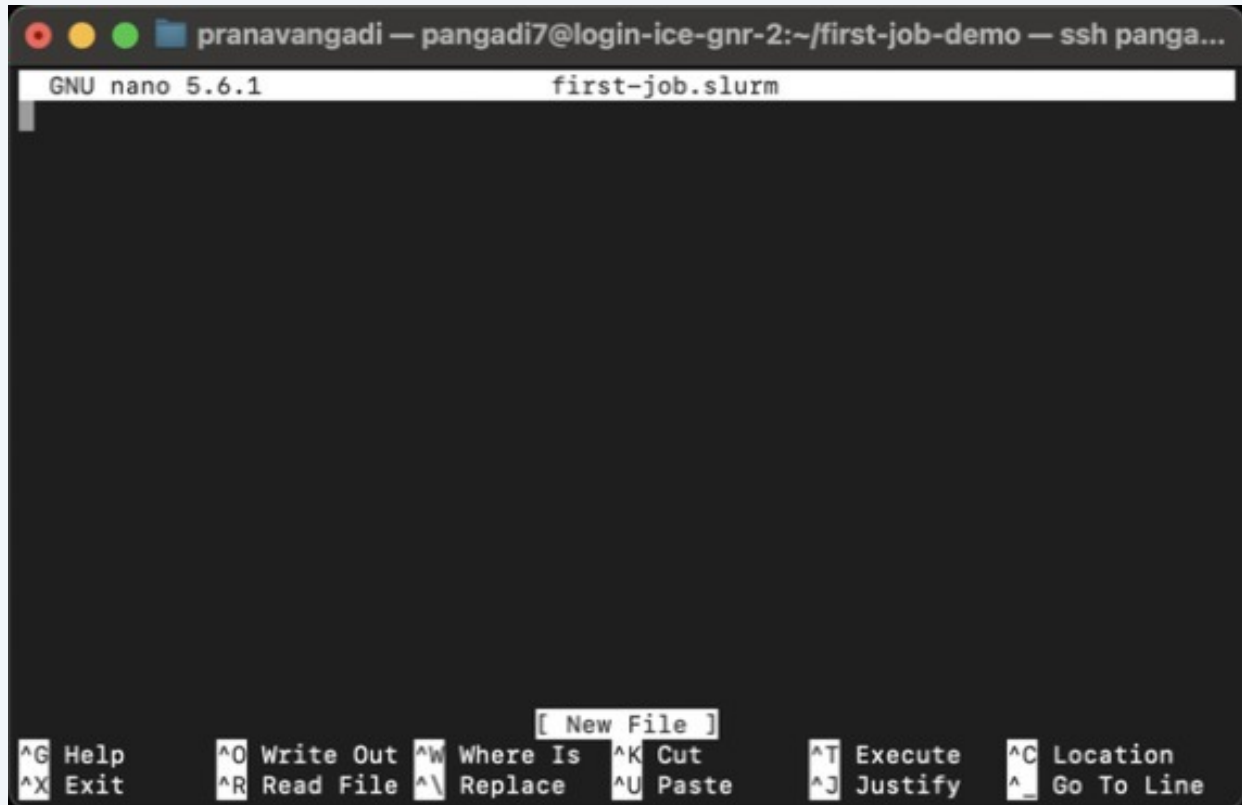
#####
# Welcome to GT Instructional Cluster #
#####

If you require assistance with this system, please contact your
course instructor or teaching assistant (TA).
[[pangadi7@login-ice-gnr-2 ~]$ mkdir -p ~/first-job-demo
[[pangadi7@login-ice-gnr-2 ~]$ cd ~/first-job-demo
[[pangadi7@login-ice-gnr-2 first-job-demo]$ pwd
/home/hice1/pangadi7/first-job-demo
[[pangadi7@login-ice-gnr-2 first-job-demo]$
```

Create the Job Script

4

Open a new Slurm script in Nano
Type "**nano first-job.slurm** Return"
It will open a window as shown below.



```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
GNU nano 5.6.1 first-job.slurm

[ New File ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

5

Paste and Save the Slurm script as first-job.slurm
Paste script then Type "**Ctrl + O Enter** "

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
GNU nano 5.6.1 first-job.slurm Modified
#!/bin/bash

#SBATCH --job-name=first-job
#SBATCH --output=first-job-%j.out
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G

echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"

sleep 60

echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
GNU nano 5.6.1 first-job.slurm Modified
#!/bin/bash

#SBATCH --job-name=first-job
#SBATCH --output=first-job-%j.out
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G

echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"

sleep 60

echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."

File Name to Write: first-job.slurm
^G Help      M-D DOS Format  M-A Append    M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend    ^T Browse
```

6

Exit Nano After Saving
Press **Ctrl + X** to exit

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
GNU nano 5.6.1 first-job.slurm
#!/bin/bash

#SBATCH --job-name=first-job
#SBATCH --output=first-job-%j.out
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G

echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"

sleep 60

echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."

[ Wrote 18 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```



Tip!

Display the file to make sure it was saved correctly:

cat first-job.slurm

Submit the Job

7

Submit the job to Slurm
Type "**sbatch first-job.slurm Return**"

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...  
/home/hice1/pangadi7/first-job-demo  
[[pangadi7@login-ice-gnr-2 first-job-demo]$ nano first-job.slurm  
[[pangadi7@login-ice-gnr-2 first-job-demo]$ cat first-job.slurm  
#!/bin/bash  
  
#SBATCH --job-name=first-job  
#SBATCH --output=first-job-%j.out  
#SBATCH --time=00:05:00  
#SBATCH --ntasks=1  
#SBATCH --cpus-per-task=1  
#SBATCH --mem=1G  
  
echo "Hello from PACE ICE!"  
echo "Job ID: $SLURM_JOB_ID"  
echo "Running on: $(hostname)"  
echo "Started at: $(date)"  
  
sleep 60  
  
echo "Finished at: $(date)"  
echo "Your first Slurm job completed successfully."  
[[pangadi7@login-ice-gnr-2 first-job-demo]$ sbatch first-job.slurm  
Submitted batch job 5554670  
[pangadi7@login-ice-gnr-2 first-job-demo]$
```

8 This number is your job ID. Your number will be different!

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...  
/home/hice1/pangadi7/first-job-demo  
[[pangadi7@login-ice-gnr-2 first-job-demo]$ nano first-job.slurm  
[[pangadi7@login-ice-gnr-2 first-job-demo]$ cat first-job.slurm  
#!/bin/bash  
  
#SBATCH --job-name=first-job  
#SBATCH --output=first-job-%j.out  
#SBATCH --time=00:05:00  
#SBATCH --ntasks=1  
#SBATCH --cpus-per-task=1  
#SBATCH --mem=1G  
  
echo "Hello from PACE ICE!"  
echo "Job ID: $SLURM_JOB_ID"  
echo "Running on: $(hostname)"  
echo "Started at: $(date)"  
  
sleep 60  
  
echo "Finished at: $(date)"  
echo "Your first Slurm job completed successfully."  
[[pangadi7@login-ice-gnr-2 first-job-demo]$ sbatch first-job.slurm  
Submitted batch job 5554676  
[pangadi7@login-ice-gnr-2 first-job-demo]$
```

Monitor Job Status

9 Check your job in the queue

Type **"squeue -u "\$USER" Return"**

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...

#SBATCH --job-name=first-job
#SBATCH --output=first-job-%j.out
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G

echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"

sleep 60

echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."
[[pangadi7@login-ice-gnr-2 first-job-demo]$ sbatch first-job.slurm
Submitted batch job 5554670
[[pangadi7@login-ice-gnr-2 first-job-demo]$ squeue -u "$USER"
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
      5554670    coc-cpu first-jo pangadi7  R        0:22      1 atl11-1-03-004-5-1
[[pangadi7@login-ice-gnr-2 first-job-demo]$
```

10 Check your job by its ID

Type **"squeue -j 5554670 Return"**

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...

#SBATCH --mem=1G

echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"

sleep 60

echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."
[[pangadi7@login-ice-gnr-2 first-job-demo]$ sbatch first-job.slurm
Submitted batch job 5554670
[[pangadi7@login-ice-gnr-2 first-job-demo]$ squeue -u "$USER"
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
      5554670    coc-cpu first-jo pangadi7  R        0:22      1 atl11-1-03-004-5-1
[[pangadi7@login-ice-gnr-2 first-job-demo]$ squeue -j 5554670
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
      5554670    coc-cpu first-jo pangadi7  R        0:43      1 atl11-1-03-004-5-1
[[pangadi7@login-ice-gnr-2 first-job-demo]$
```



Tip!

- R means the job is running.
- PD means the job is waiting to run.
- Wait about one minute and check again. If the job no longer appears, it has left the queue.

Verify Job Output

11

Find the Slurm output file.
Type "**ls -lh Return**"

```

pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
sleep 60

echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."
[[pangadi7@login-ice-gnr-2 first-job-demo]$ sbatch first-job.slurm
Submitted batch job 5554670
[[pangadi7@login-ice-gnr-2 first-job-demo]$ squeue -u "$USER"
          JOBID PARTITION    NAME    USER  ST       TIME  NODES NODELIST(REAS
SON)
          5554670    coc-cpu first-jo pangadi7  R         0:22      1 atl11-1-03-00
4-5-1
[[pangadi7@login-ice-gnr-2 first-job-demo]$ squeue -j 5554670
          JOBID PARTITION    NAME    USER  ST       TIME  NODES NODELIST(REAS
SON)
          5554670    coc-cpu first-jo pangadi7  R         0:43      1 atl11-1-03-00
4-5-1
[[pangadi7@login-ice-gnr-2 first-job-demo]$ squeue -j 5554670
          JOBID PARTITION    NAME    USER  ST       TIME  NODES NODELIST(REAS
SON)
[[pangadi7@login-ice-gnr-2 first-job-demo]$ ls -lh
total 8.0K
-rw-r--r--. 1 pangadi7 gtperson 855 Jul 31 19:30 first-job-5554670.out
-rw-r--r--. 1 pangadi7 gtperson 369 Jul 31 19:29 first-job.slurm
[[pangadi7@login-ice-gnr-2 first-job-demo]$

```

12 Read the job's output.

Type "**cat first-job-5554670.out Return**"

A successful job should end with "Your first Slurm job completed successfully"

```
pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
Account: coc
Job name: first-job
Partition: coc-cpu
QOS: coc-ice
-----
Hello from PACE ICE!
Job ID: 5554670
Running on: atl1-1-03-004-5-1.pace.gatech.edu
Started at: Fri Jul 31 07:29:53 PM EDT 2026
Finished at: Fri Jul 31 07:30:53 PM EDT 2026
Your first Slurm job completed successfully.
-----
Begin Slurm Epilog: Jul-31-2026 19:30:53
Job ID: 5554670
User ID: pangadi7
Account: coc
Job name: first-job
Resources: cpu=1,mem=1G,node=1
Rsrc Used: cput=00:01:00,vmem=0,walltime=00:01:00,mem=2516K,energy_used=0
Partition: coc-cpu
QOS: coc-ice
Nodes: atl1-1-03-004-5-1
-----
[pangadi7@login-ice-gnr-2 first-job-demo]$
```